

An Approach for EEG Data Augmentation Based on Deep Convolutional Generative Adversarial Network

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Abstract—High-quality Electroencephalogram (EEG) signals are crucial for BCI applications. In practice, processing and analysis of EEG are often hindered due to either data scarcity or data imbalance, which would lower the performance of EEG-based classification. In this work, we implemented a data augmentation approach based on Deep Convolutional Generative Adversarial Network (DCGAN), which employed a generator and a discriminator to compete with each other to gain high-quality data for augmentation. In the approach, one-dimensional (1D) convolution layers were used by the discriminator to extract time features of raw signals, and 1D transposed convolutional layers were adopted by the generator to synthesize new data. The quality of synthesized data was analyzed in time domain, and as well in both frequency and time-frequency domains by using Fast Fourier Transform and Continuous Wavelet Transform. The experimental results showed that the synthesized data have very similar temporal and spectral features to the raw data, which confirmed the effectiveness of our proposed method in synthesizing useable data from limited original signals with a concise structure and fewer network parameters. This work provides a possible deep learning method for EEG data augmentation with a lot of application potentials in BCI areas.

Index Terms—Data Augmentation, EEG, Brain-computer interface (BCI), Generative Adversarial Network (GAN)

I. INTRODUCTION

Brain-Computer Interface (BCI) is a novel technology that enables people to directly interact with external environment with brain waves, for example, it could decode patients' motor intention into control comments for external equipment, like prostheses and exoskeletons [1]. The benefit of BCI over conventional therapeutic approaches is users do not need to

move their limbs for device control, which is a breakthrough in the fields of rehabilitation engineering and bio-robotics. To precisely manipulate prostheses or exoskeletons with BCI, we need to obtain high-quality Electroencephalogram (EEG) data, which can be recorded invasively or non-invasively and reflect brain activities from neuronal potential changes. Since EEG has multiple merits like high temporal resolution and stability in time domain, thus it can be used in various tasks like classification.

In the past decades, many studies have placed tremendous concentration and importance on EEG task classifications [2], and classifications of Motor Imaginary (MI) [3, 4] and Emotional Recognition (ER) [5] are eye-catching works in neural engineering since they can aid in psychological diagnoses and controlling external devices for patients with depression and paralyzed people. For task classifications and predictions, Deep Learning (DL) [6] is a mainstream methodology that is widely used in Computer Vision (CV) and Nature Language Processing (NLP) areas. However, the DL techniques require a huge amount of data and EEG usually suffers data scarcity and imbalance due to expensive acquisition procedures, and massive data are not publicly available because of privacy issues, thus data augmentation methods are all we need.

In the field of CV, it is a common way that researchers randomly resize or rotate the image to obtain new pictures, but one cannot directly do such geometric transformations on EEG data to enlarge the data size, since the time features and spatial features will be undermined. Some studies added Gaussian noise or combine segmented and original data to achieve data augmentations. Wang *et al.* added Gaussian white noise into their original training data to gain new samples for ER classification [7]. Kuanar *et al.* transformed EEG data from time series into the format of images and carried various Gaussian noise to the transformed data [8]. Salama *et al.* generated noisy EEG signals by adding Gaussian noise with zero mean and unified variance [9]. However, adding a small amount of noise to the original EEG data may easily mislead the classifiers, and more and more works tend to implement Generative Adversarial Networks (GANs) to produce new and realistic data. GAN is an unsupervised and generative DL algorithm and was firstly proposed by Goodfellow *et al.* [10], which is made up of two neural networks competing with each other and can capture and copy features within the dataset. It is virally popular in the CV areas to synthesize non-existent images or transfer styles of images by adversarial training. Then Radford *et al.* introduced a novel GAN framework named Deep Generative Adversarial Network (DCGAN), which bridges the gap between the Convolutional Neural Network (CNN) and the unsupervised

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model and is capable of learning hierarchical representations from raw data [11].

Regarding the EEG data generation, some researchers made efforts to apply GAN-based models in data augmentation. Song *et al.* proposed a cross-subject EEG classification framework with common spatial GAN (CS-GAN), which gained high-quality data and significantly improved MI classification accuracy [12]. Corley *et al.* stated the cost of hardware is increasingly expensive when multi-channel EEG is being recorded simultaneously, and they used GANs to output super spatial resolution EEG with low spatial resolution input [13]. Wasserstein GAN (WGAN) is another GAN model with more stability in learning and which can get rid of model collapse, and some researchers implemented WGAN in EEG data generation to enhance the performance of EEG-based classification [14]. However, most existing models have complicated structures and enormous quantity of parameters to pursue stability in training and diversity in generation, which may heavily burden the learning speed of networks.

In this paper, we utilized a concise Deep Convolutional GAN (DCGAN) structure to generate real-like EEG data for better classification performance, and the main contributions are summarized as follows:

- We designed a novel DCGAN framework for EEG augmentation, and implemented one-dimensional (1D) instead of two-dimensional (2D) convolutional layers to better extract features of time series.
- We evaluated the generated EEG signals in both time and frequency domains and plotted the power spectral map in multiple channels, based on which the performance of the generator on synthesizing time and spectral features of complicated real EEG signals was analyzed.

II. METHODS AND MATERIALS

A. Generative Adversarial Network

Generative Adversarial Network is a deep-learning method proposed by Goodfellow *et al.* in 2014, which is inspired by game theory and was firstly used in CV to generate fake images. The network consists of two modules: a generator and a discriminator. The Generator is dedicated to generating fake data from the random input to fool the discriminator, and the discriminator tries to classify fake data and real input. After multiple rounds of adversarial training, the generator could produce fake data that the discriminator cannot accurately distinguish anymore, which we called Nash equilibrium. The whole training process can be seen as a game between a counterfeiter and a police. The counterfeiter is committed to making fake money to deceive the police, while the police should distinguish between real and fake money as accurately as possible. During the training process, the generator is fed into random noise and outputs the data that is in real data shape, and the discriminator is fed into both real data and generated data.

To demonstrate the adversarial learning procedure clearly, the optimization goal could be formularized below:

$$\min_{\theta_g} \max_{\theta_r} V(G, D) = E_{x_r \sim p_r} [\log(D(x_r))] + E_{x_g \sim p_g} [\log(1 - D(x_g))], \quad (1)$$

where θ_d , θ_g , x_r and x_g are the discriminator parameters, generator parameters, real data and synthetic data respectively, $D(x)$ returns the probability of x belonging to real (p_r) or generated data (p_g) distributions. Specifically, the input data to the generator are random noises, it can be stated as below:

$$x_g = G(z) \quad (2)$$

In the adversarial learning procedure, the network trains a generator and a discriminator simultaneously, and the generator and discriminator compete against each other to maximize the $\log(D(x_r))$ and minimize the $\log(1 - D(x_g))$.

B. Dataset Description and preprocessing

We collected the data of three subjects from a public dataset named EEG Motor Movement/Imagery dataset from Physionet [15], which consists of over two-minute EEG recordings and obtained from 109 volunteers. In this dataset, 64 channels of EEG signals were simultaneously recorded at 160 samples per second with the BCI2000 system (<http://www.bci2000.org>) based on the international 10-10 system [16]. Each subject performed 14 experiments: two one-minute baseline runs (one with eyes open and the other with eyes closed), three two-minute runs of four tasks with the orders shown by a screen: opening and closing the corresponding hands, imaginary opening and closing the corresponding hands, opening and closing the corresponding feet, imaginary opening and closing the corresponding feet.

For data preprocessing, we used only the first 19900 data points, normalizing data in the range of 0-1 to avoid gradients exploding and gradients vanishing in the training process. For performance evaluation, we implemented a process of inversed-normalization to the output data from the generator to precisely compare the fake data and real data.

C. Architecture and Hyper-parameters

The architecture of the network is illustrated in Figure 1. We chose convolutional blocks in the discriminator to extract features and transposed convolutional blocks in the generator to synthesize EEG data.

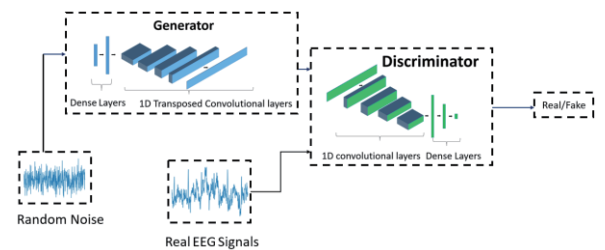


Figure 1. Deep Convolutional GAN (DCGAN) Architecture.

Considering the placement of EEG electrodes can be seen as an image, some studies used 2D convolutional layers to extract features, where 64 EEG channels are treated as an 8×8 map and convolutional operations are subsequently performed on this map [17]. However, in this way the whole architecture was inevitably limited by map size and could not compose

many convolutional layers. In this study, therefore, we utilized 1D convolutional layers to extract features followed by batch normalization and Parametric Rectified Linear Unit (PReLU) as activation function, and implemented 1D Transposed convolution blocks to generate fake EEG signals.

(a) Generator

The generator was employed to synthesize the fake EEG data similar to the real EEG data that related to several motor imageries. It is worth noting that our input random Gaussian noise had 64 channels and 19900 data points like the real data shape. Therefore in the generator, we used two dense blocks and each of them consists of a dense layer and a PReLU activation function, transforming random input into an appropriate shape to match the following operations. Then, we used five transposed convolution blocks with different kernel sizes and strides, each block has a 1D transposed convolution layer, a batch-normalization layer, and an activation function. Specifically, the activation function in the first four blocks is PReLU while in the last block is sigmoid.

(b) Discriminator

The discriminator composes five 1D convolution blocks, each block consists of a 1D convolutional layer, a batch normalization layer, and a PReLU activation function. However, the neural networks are easily over-fit and the dropout regularization method is widely used in the deep-learning field to alleviate this problem, so we added a dropout layer in each convolutional block to randomly drop out neurons with a certain probability. Apart from convolution layers, we still implemented two dense layers to judge the reality of the input data. Like the conventional generative adversarial network, the original real data and generated fake data were inputted into the discriminator during the training procedure.

(c) Hyper-parameters

The hyper-parameters of the network play a crucial role in generating real-like signals, and the generator parameters and the discriminator parameters are shown in Table I and Table II, respectively. The discriminator has a dropout rate of 0.5 in each convolutional block, the batch size is 16, and the adversarial loss is binary cross-entropy loss. The optimizer for both generator and discriminator is Adam with a learning rate $\alpha=10^{-3}$ and beta values $\beta_1=0.5$, and $\beta_2=0.99$. After 500 training epochs, we judged the performance of generated data.

III. RESULTS

In this work, similar trends and results were obtained across all the three subjects. Due to limited paper length, the results of only one subject are illustrated. The discussion and conclusion are still based on all the tested subjects.

A. Time domain analysis

Time domain analysis can visualize the characteristics of signals over time. After collecting the generated data, it is important to evaluate the resemblance between generated data and real data over 64 channels, ensuring that the generated EEG data embody similar temporal patterns as real data for some subsequent operations like event-related potential analysis, source localization, and functional connectivity analysis. We analyzed each channel of the generated data and

TABLE I. GENERATOR PARAMETERS

Layers	Input	Output	Kernel	Stride
Dense	19900	2816		
	2816	2476		
Transposed Convolution	2476	4970	(1,20)	4
	4970	9948	(1,10)	2
	9948	19899	(1,5)	2
	19899	19900	(1,2)	2
	19900	19900	(1,1)	4

TABLE II. DISCRIMINATOR PARAMETERS

Layers	Input	Output	Kernel	Stride
Convolution	19900	4971	(1, 20)	4
	4971	2481	(1,10)	2
	2481	1239	(1,5)	2
	1239	619	(1,2)	2
	619	155	(1,1)	4
Dense	39680	2816		
	2816	1		

found that every channel has similar temporal features, and the generated data coincided with real data in the whole time range. We selected one channel to plot the first 1000 points of the real and generated data with inversed-normalization in Figure 2, and we used the same channel in the following analysis. As can be seen, the generated data are perfectly close to the real data, and spikes of the original data could be completely synthesized in spite of the higher or lower deviated values.

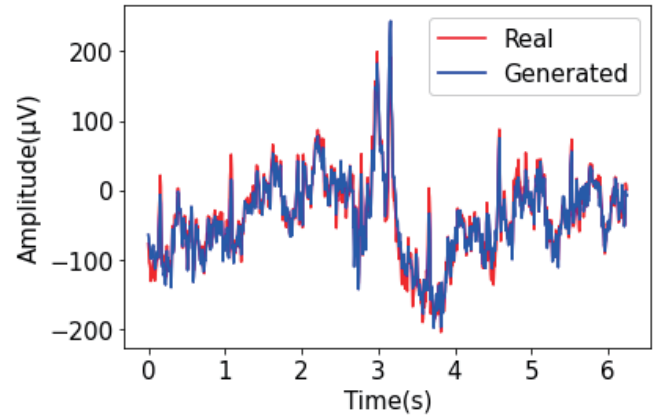


Figure 2. Real (red) and Generated (blue) signals in time domain

B. Frequency domain analysis

For frequency domain analysis, we implemented Fast Fourier transform (FFT) to analyze the frequency features of the first 1000 data points for each channel. Figure 3 (a) and (b) show the real data and generated data of the same channel as shown in Figure 2 in frequency domain. It is noticeable that the curves of real data and generated data have the same trends in plots and the frequencies of the first 1000 data points are mainly in 0-10Hz for both real data and fake data.

C. Spectral map

Joint Time-Frequency Analysis (JTFA) can provide the joint distribution information between the time domain and frequency domain. The core idea of JTFA is to design a joint function of both time and frequency and use it to describe the

energy density or intensity of the signal at variable time and frequency. In this study, we operated JTFA for the real data and generated data by Continuous Wavelet Transform (CWT), and analyzed all 64 channels and found that all the channels have similar patterns in spectral maps. Figures 4 (a) and (b) illustrate typical spectral maps for the same channel as shown in Figure 2. It is noticeable that the generated signals fit the original signals quite well, with power ranging from 0-2.5 Hz and 0-1 s.

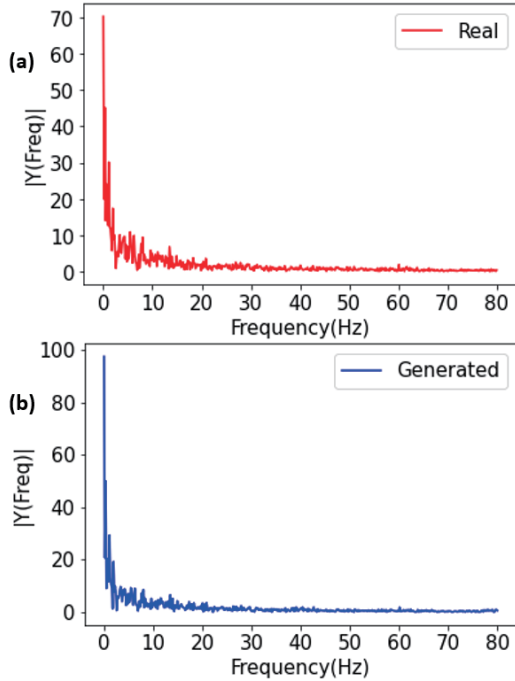


Figure 3. (a) Real and (b) Generated signals in frequency domain.

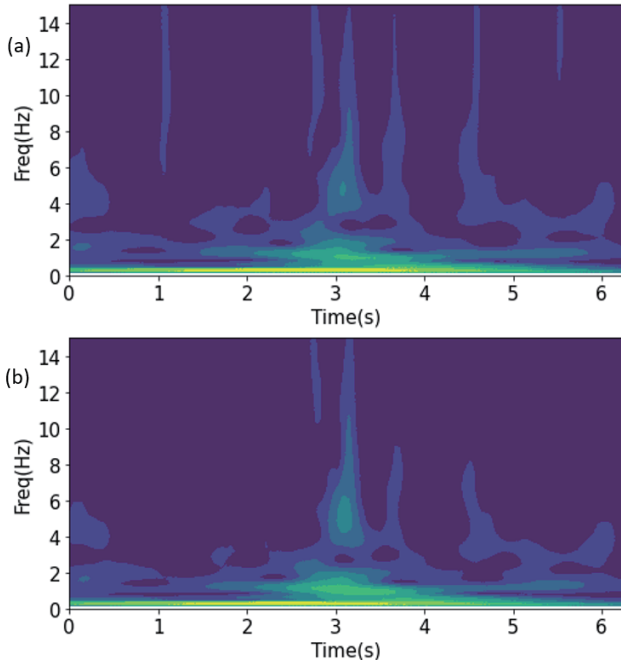


Figure 4. Continuous wavelet transform for the (a) Real data and (b) Generated data.

IV. DISCUSSION

Data imbalance and data scarcity are the main problems in EEG classification and prediction, which limit the accuracy of clinical applications. The Deep Convolutional Generative Adversarial Network (DCGAN) was utilized to solve these problems. We highlighted the data normalization techniques and the 1D convolutional layers to extract features of time series. The results demonstrated that 1D convolutions can effectively extract EEG patterns without undermining local features and are not limited by map size like some 2D convolutions [17]. Besides, normalizing data into a fixed range could effectively alleviate gradient exploding or gradient vanishing, while we still have to restore the scale of the generated data in the evaluation stage by inversed data normalization to compare with the real data.

In some pilot works on EEG data augmentation, Gaussian noise was added to EEG but this way may damage local features [18]. Subsequently, with the emergence of GAN, many works applied this technique in EEG data generation. Compared with the works which synthesized non-existent EEG data and analyzed data only in time domain [19], we analyzed the generated data over time, frequency, and time-frequency domains. And the results confirmed robustness of the proposed network and showed that generated data had satisfying quality and temporal resolution, which means the generated data could perfectly mimic the real data. However, even though GAN-based models have made huge successes in many fields, this method still owns some shortcomings. It is time-consuming and computing resource-consuming to train an adversarial network, especially for long time-series data with multiple convolutional blocks. The discriminator during the competition procedure may behave distinguishable in the very beginning and it easily judges fake and real data, which may cause the generator could not synthesize high-quality data and the whole model to collapse. A possible solution is reiterating the discriminator training several times while the generator is only trained one time in an epoch, which may make the adversarial training process more stable and robust.

Although we generated high-quality EEG signals with only five convolutional blocks, considering the complicated EEG data structure and a huge amount of data points, it is suggested to try deeper layers with residual connections to get super-resolution data in future works. Besides, future work could consider the performance of GAN-based data augmentation methods in actual EEG classifications and predictions, and judge the reality and reliability of generated data, exploring whether these methods could solve data imbalance and scarcity problems and get promising results. Additionally, the method for evaluating the quality of generated data still needs to be explored. There are no such widely accepted metrics to judge the quality of generated EEG, and further research could find a method to objectively evaluate the generated data in mathematical and statistical ways and judge for their usability and reliability.

V. CONCLUSION

Data imbalance and data scarcity hinder the accuracy of EEG classification and BCI applications, an efficient and reliable approach to enlarge data size may solve these problems. This paper explored the EEG data augmentation

method based on Deep Convolutional Generative Adversarial Network (DCGAN) with 1D convolutional layers. After observing the generated data, we found they had similar temporal and spectral features to real EEG samples. The results indicate that the generated data by DCGAN have high quality and could be a reliable source for EEG analysis.

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